

SOFTWARE ENGINEERING MINI PROJECT
RETAIL INVENTORY AND STOCK MANAGEMENT
SYSTEM
TEST PLAN

Deliverable Part-1 - TEAM E10

Team Members

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1. Introduction

1.1 Purpose

This Test Plan defines the verification and validation activities for the Retail Inventory and Stock Management System (RISMS). It is aligned with the Phase-1 Software Requirements Specification (SRS) and the Software Architecture & Design Specification. The plan covers functional, verifiable functional, non-functional, security and persistence requirements and provides traceability from requirements and use cases to test cases.

The plan follows an IEEE-style test-plan organization and uses the project terminology and requirement IDs already established in the SRS and design document. No new functional requirements are introduced.

1.2 Scope

RISMS is a standalone, menu-driven C++ application for small-retail inventory operations. Testing covers product management, search and viewing, stock adjustment, stock availability, stock movement history, inventory listing, inventory summary, authentication/authorization and file-based persistence.

- In scope: UC01–UC11, FR01–FR14, VFR01–VFR09, NFR01–NFR08 and SEC-01–SEC-04.
- In scope: design safeguards that implement the SRS security and integrity requirements, including session gating, role checks, password hashing, lockout, atomic writes, append-only ledger handling and owner-only data-directory permissions.
- Out of scope: graphical/web interface, online shopping, payment processing, supplier/purchase-order management, external database, networking, barcode hardware and external retail-system integration.

1.3 References

- Retail Inventory and Stock Management System — Software Requirements Specification (SRS), Team E10, Phase 1..
- The SRS defines FR01–FR14, VFR01–VFR09, NFR01–NFR08, SEC-01–SEC-04 and UC01–UC11.
- The design document defines the three-layer architecture, repository pattern, security architecture, error handling and key algorithms.

1.4 Definitions and Abbreviations

Term	Meaning
RISMS	Retail Inventory and Stock Management System
SRS	Software Requirements Specification
FR	Functional Requirement
VFR	Verifiable Functional Requirement
NFR	Non-Functional Requirement
SEC	Security Requirement
UC	Use Case
RTM	Requirements Traceability Matrix
Ledger	Append-only stock movement history stored in stock_ledger.csv

Atomic write	Write to a temporary file and replace the original in one filesystem operation
Stock-in	Increase of a product's quantity
Stock-out	Decrease of a product's quantity

2. Test Plan Identification

Item	Baseline / Decision
System under test	RISMS C++17 console application
Primary interfaces	Console menu and C++ class/module interfaces; there is no network API
Deployment	Single store PC; Windows 10+ or Linux; local data/ directory
Persistence	products.csv, stock_ledger.csv, users.dat, config.ini and error.log
Actors	Administrator and Inventory Staff
Authentication	Both actors authenticate; role-specific access follows UC definitions
Test data	Controlled product records, valid/invalid inputs, stock quantities, user accounts and configured thresholds
Test levels	Unit/module, integration, system and inspection/review
Execution evidence	Console output, persisted files, ledger entries, error.log entries and requirement/test results

3. Test Items

The following software items are testable components or interfaces identified by the architecture and design baseline. The test plan exercises them through unit/module tests, integration tests and end-to-end console tests as appropriate.

Test Item	Main Responsibility	Related Requirements
ConsoleUI / MenuController	Input parsing, prompts, menu dispatch and session gate	NFR01, NFR02, FR01, SEC-01
AuthService / UserRepository	Authentication, password verification, session and lockout	FR01, SEC-01
ProductService / Validator	Add, update, remove, search, list and validation	FR02–FR06, FR11, VFR01–VFR04, SEC-02, NFR02
StockService	Stock-in/out, quantity arithmetic and availability validation	FR07–FR09, VFR05–VFR07, SEC-03, SEC-04

ReportService / ConfigStore	History retrieval, inventory summary and threshold handling	FR10, FR12, VFR09
ProductRepository / FileStore	Product persistence, loading and atomic writes	FR13, FR14, VFR08, NFR04
LedgerRepository	Append-only movement history and retrieval	FR09, FR10, VFR07, SEC-04
Error handling	Typed errors, rollback, corruption handling and logging	NFR02, NFR03, NFR04, SEC-04
Complete RISMS executable	Integrated local inventory application	FR01–FR14, VFR01–VFR09, NFR01–NFR08, SEC-01–SEC-04

4. Features to be Tested

Feature	Traceability	Verification Focus
4.1 User Access and Authentication	UC01 / FR01 / SEC-01	Valid login, invalid credentials, session gate, logout and loading saved inventory.
4.2 Product Management	UC02–UC04 / FR02–FR04 / VFR01–VFR02	Add, update and remove products; unique IDs; required fields; role restrictions.
4.3 Product Search and Viewing	UC05–UC06 / FR05–FR06 / VFR03–VFR04	Search by ID/name, valid result, not-found handling and product details.
4.4 Stock Adjustment	UC07 / FR07 / VFR05–VFR06 / SEC-03	Stock-in arithmetic, stock-out arithmetic, validation and insufficient-stock rejection.
4.5 Stock History	UC07/UC09 / FR09–FR10 / VFR07 / SEC-04	One successful adjustment produces one ledger entry; history is retrievable and retained.
4.6 Inventory Listing and Summary	UC10–UC11 / FR11–FR12 / VFR09	Inventory listing, totals and low-stock identification using configured threshold.
4.7 Persistence and Recovery	UC01 / FR13–FR14 / VFR08 / NFR03–NFR04	File storage, restart persistence, atomic save behavior and recovery from file errors/corrupt lines.
4.8 Usability and Input Validation	NFR01–NFR02	Clear menu prompts, actionable validation messages and safe numeric parsing.
4.9 Maintainability	NFR05	Inspection that presentation, services and repositories remain separated and independently testable.
4.10 Portability and Local Operation	NFR06 / NFR08	Build/run on compatible local C++ environment without external database or network service.
4.11 Performance	NFR07	Basic operations execute without unnecessary processing delays under normal usage; no numeric threshold is invented because the SRS specifies none.

5. Test Approach

Testing combines requirements-based testing, use-case/system testing, module/integration testing and inspection of non-functional design properties. The architecture explicitly supports module testing because services receive repository interfaces and can be exercised with in-memory fakes.

- Functional testing verifies each user-visible operation and its expected success and failure behavior.
- VFR testing uses concrete input/output conditions such as duplicate rejection, insufficient-stock rejection, persistence after restart and low-stock threshold behavior.
- Non-functional testing verifies usability, validation, reliability, persistence, maintainability, portability, normal-use performance and local-only operation.
- Integration testing verifies coordination between services, repositories, ProductRepository/FileStore and LedgerRepository.
- Negative testing deliberately supplies invalid, incomplete, duplicate, unauthorized and excessive values.
- Regression testing repeats affected test cases after fixes or changes to shared services, repositories or validation rules.

5.1 Security Validation

Security validation is directly tied to SEC-01 through SEC-04 in the SRS. The design document also defines additional safeguards—salted password hashing, constant-time comparison, three-attempt/60-second lockout, owner-only data-directory permissions, append-only ledger behavior and atomic file writes. These safeguards are validated as implementation controls supporting the SRS requirements; they are not treated as new SRS requirement IDs.

Security Area	Validation	Requirement
Authentication / access control	Unauthenticated users cannot execute protected inventory operations; valid Administrator and Inventory Staff accounts can enter according to the SRS actor model.	SEC-01
Role authorization	Administrator-only UC04 and UC11 are blocked for Inventory Staff; Inventory Staff-only UC07 and UC09 are not made available to Administrator where the SRS actor matrix specifies otherwise.	SEC-02 / actor access matrix
Stock integrity	Invalid quantity, non-positive quantity and stock-out beyond available quantity are rejected before stored quantity is changed.	SEC-03
Ledger integrity	Successful stock changes are appended to the ledger; no update/delete interface is exposed; failed persistence does not leave a false successful movement.	SEC-04
Authentication safeguards	Wrong credentials are handled generically; three consecutive failures trigger the configured 60-second lockout; successful login resets the failure count.	SEC-01 supporting control
File/storage safeguards	Products/users are written atomically; malformed stored lines are skipped and logged; data directory is restricted to the owning OS account.	SEC-04 / NFR04 supporting controls

Audit logging	Login success/failure, lockout, logout and I/O errors are written to error.log and not displayed as raw internal details to users.	SEC-01 / SEC-04 supporting control
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6. Entry and Exit Criteria

6.1 Entry Criteria

- SRS and Software Architecture & Design Specification are baselined for testing.
- A compilable RISMS C++17 build is available.
- Test data and user accounts for Administrator and Inventory Staff are available.
- The local data directory and configuration can be initialized/reset between tests.
- Required test cases and expected results have been reviewed against the SRS and design.

6.2 Exit Criteria

- All planned test cases have been executed or an approved reason for deferral is documented.
- All SEC-01–SEC-04 tests have been executed.
- All FR01–FR14 and VFR01–VFR09 requirements have at least one verification mapping and execution result.
- No unresolved Critical/High defect remains that prevents a core inventory operation or violates a security requirement.
- Failed tests have corresponding defect records or documented corrective action.
- The final RTM/test-result set is complete.

7. Pass/Fail Criteria

Result	Criterion
PASS	Actual result satisfies the expected result and the linked requirement is demonstrated.
FAIL	Actual result violates the expected result, produces incorrect data, bypasses authorization, loses history or otherwise fails the linked requirement.
BLOCKED	Execution cannot continue because a prerequisite, build, environment or dependent defect is unavailable.
NOT RUN	Test is planned but has not yet been executed.

8. Test Suspension and Resumption

- Suspend testing if the application cannot start, data files cannot be initialized, or a blocking defect prevents meaningful execution of dependent cases.
- Security validation is suspended only when the build or authentication subsystem is unavailable; independent security/design inspections may continue.

- After the blocking issue is fixed, rerun the failed test and all directly affected regression cases before resuming dependent tests.

9. Test Environment

Environment Item	Required Configuration
Operating system	Windows 10+ or Linux, consistent with the design deployment view
Compiler/build	C++17; g++/MSVC Community and/or the project CMake target
Application	risms / risms.exe
Storage	Local data/ directory containing products.csv, stock_ledger.csv, users.dat, config.ini, error.log
Input	Keyboard
Output	Console display
Network	Not required; NFR08 is validated by running without network/external services
Test accounts	At least one Administrator and one Inventory Staff account
Test data	Products with normal, zero, boundary, low-stock and insufficient-stock quantities

10. Test Deliverables

- Approved Test Plan.
- Test case specification and execution results.
- Requirements Traceability Matrix (RTM).
- Defect/issue records for failed tests.
- Test evidence: console output, file contents, ledger entries and error.log evidence where applicable.
- Final test summary/result status.

11. Test Schedule

Activity	When / Condition	Output
Test planning and review	Before execution	Baselined Test Plan
Unit/module verification	After module implementation	Module test results
Integration testing	After service/repository integration	Integration results
System/use-case testing	After executable build is stable	UC01–UC11 results
Security validation	During system testing and regression	SEC-01–SEC-04 results
Non-functional testing	After core functions are stable	NFR results

Regression testing	After fixes affecting tested components	Regression results
Final traceability review	Before submission	Completed RTM and test summary

12. Responsibilities

Role	Responsibility
Test team	Prepare data, execute test cases, capture evidence, record results and report defects.
Developers	Provide build, correct defects, support module/integration tests and maintain implementation consistency with the SRS/design.
Test lead / team coordinator	Review traceability, monitor coverage and approve test completion.
Project team	Review final Test Plan, RTM, test results and defect status before submission.

13. Risks and Mitigation

Risk	Impact	Mitigation
File corruption or I/O failure	Incorrect or lost inventory data	Test atomic writes, rollback behavior, restart persistence and DATA_CORRUPT handling.
Authorization bypass	Unauthorized inventory modification	Execute role-based and unauthenticated-access tests for protected use cases.
Incorrect stock arithmetic	Incorrect inventory quantities	Use boundary and insufficient-stock tests plus ledger verification.
Requirement/design mismatch	Inconsistent deliverables	Maintain RTM against the exact SRS IDs and design mappings.
No numeric performance target in SRS	Ambiguous performance pass criterion	Validate absence of unnecessary delays under normal usage without inventing a numeric threshold.
Environment-specific file permissions	Security control behaves differently across OS	Inspect and test data-directory permissions on the supported OS environment.

14. Requirements Traceability Matrix

The following matrix maps every SRS requirement to at least one test case. The requirement IDs and use-case relationships are retained from the SRS; the test cases are the verification activities defined in this Test Plan.

Req. ID	Requirement	Use Case	Test Case(s)
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FR01	User Access	UC01	TC-01, TC-13
FR02	Add Product	UC02	TC-02, TC-03
FR03	Modify Product	UC03	TC-04
FR04	Remove Product	UC04	TC-05
FR05	Search Product	UC05	TC-06
FR06	View Product Details	UC06	TC-06
FR07	Adjust Stock	UC07	TC-07, TC-08
FR08	Check Stock Availability	UC08	TC-07, TC-08
FR09	Record Stock Adjustments	UC07	TC-09
FR10	View Stock Movement History	UC09	TC-09
FR11	View Inventory	UC10	TC-10
FR12	Generate Inventory Summary	UC11	TC-10
FR13	Store Inventory Data	UC02/UC07	TC-02, TC-11, TC-14
FR14	Load Stored Data	UC01	TC-01, TC-11
VFR01	Duplicate product IDs rejected	UC02	TC-03
VFR02	Invalid/incomplete product input rejected	UC02	TC-03, TC-12
VFR03	Valid search returns matching record	UC05	TC-06
VFR04	Unknown search key returns not-found	UC05	TC-06
VFR05	Stock-in updates quantity	UC07	TC-07
VFR06	Stock-out cannot exceed available quantity	UC07	TC-08
VFR07	Successful stock changes are logged	UC07	TC-09, TC-14
VFR08	Saved records remain after restart	UC01	TC-01, TC-11
VFR09	Low-stock threshold is respected	UC11	TC-10
NFR01	Usability	—	TC-15
NFR02	Input Validation	—	TC-03, TC-12
NFR03	Reliability	—	TC-08, TC-11, TC-14
NFR04	Data Persistence	—	TC-01, TC-11, TC-14
NFR05	Maintainability	—	TC-16
NFR06	Portability	—	TC-17
NFR07	Performance	—	TC-15
NFR08	Local Operation	—	TC-17

SEC-01	Authorized access only	UC01 / access matrix	TC-01, TC-05, TC-13
SEC-02	Restrict unauthorized product modification/removal	UC02–UC04	TC-05
SEC-03	Validate stock adjustments before update	UC07	TC-08, TC-12
SEC-04	Maintain stock movement history integrity	UC07/UC09	TC-09, TC-14

15. Test Case Specification

Seventeen test cases are included to provide coverage of functional and non-functional requirements, with dedicated security validation. This exceeds the minimum 7–10 test-case requirement and avoids introducing unsupported requirements.

ID	Title	Type	Requirement(s)	Preconditions	Steps / Test Data	Expected Result	UC
TC-01	User login and saved-data load	Functional / Security	FR01, FR14, VFR08, NFR04, SEC-01	Administrator or Inventory Staff account exists; saved files contain valid records.	Start application; enter valid credentials; observe login and loading; verify saved product/history records are available.	User reaches the inventory menu; saved records load correctly; unauthenticated access is not granted.	UC01
TC-02	Add valid product and persist	Functional	FR02, FR13	Authenticated Administrator or Inventory Staff; unique product ID available.	Select Add Product; enter valid ID, name, category, price and quantity; save; inspect inventory/data file.	Product is added once with entered values and is stored in products.csv.	UC02
TC-03	Reject duplicate and invalid product input	Functional / Validation	FR02, VFR01, VFR02, NFR02	Authenticated user; existing product ID available.	Try duplicate ID; try empty/over-length/negative/non-numeric product fields; observe validation.	Duplicate ID and invalid/incomplete input are rejected; no invalid record is stored; actionable message is shown.	UC02
TC-04	Update product details	Functional	FR03	Authenticated user; existing product.	Select Update Product; modify name/category/price; save; verify product.	Existing product details change correctly; quantity is not modified through product update.	UC03
TC-05	Remove product and verify role control	Functional / Security	FR04, SEC-02	Existing product; Administrator and Inventory Staff accounts.	As Administrator, remove a product with confirmation. As Inventory Staff, attempt the same protected operation.	Administrator can remove after confirmation; Inventory Staff is denied; unauthorized removal does not occur.	UC04
TC-06	Search and view product details	Functional	FR05, FR06, VFR03, VFR04	At least one known product exists.	Search by exact ID; search by name; search using an unknown key; open product details.	Known search returns matching record; unknown key returns not-found; selected product details display correctly.	UC05/UC06

TC-07	Stock-in and stock-out calculation	Functional	FR07, FR08, VFR05	Product with known starting quantity; Inventory Staff account.	Perform valid stock-in; check quantity; perform valid stock-out; check quantity.	Stock-in adds entered amount; stock-out subtracts entered amount; current quantity is displayed correctly.	UC07/UC08
TC-08	Reject stock-out greater than available	Functional / Security / Reliability	FR07, FR08, VFR06, NFR03, SEC-03	Product quantity known; requested stock-out greater than available.	Attempt stock-out greater than current quantity; inspect quantity and output.	Operation is rejected with insufficient-stock message; quantity remains unchanged; no successful movement is recorded.	UC07
TC-09	Record and retrieve stock movement history	Functional / Security	FR09, FR10, VFR07, SEC-04	Product exists; valid stock adjustment possible.	Perform one stock-in and one stock-out; inspect stock_ledger.csv; open stock history.	Each successful adjustment creates one ledger entry containing product ID, change, date/time and type; history is retrievable in newest-first order.	UC07/UC09
TC-10	Inventory listing and low-stock summary	Functional	FR11, FR12, VFR09	Products include quantities above and at/below configured threshold.	View inventory; run inventory summary; change/use configured threshold as permitted by test setup.	All products are listed; summary shows quantities and identifies products whose quantity is at or below the configured threshold.	UC10/UC11
TC-11	Persistence after restart	Non-functional / Functional	FR13, FR14, VFR08, NFR03, NFR04	Valid product and ledger records exist.	Add/update product and perform stock adjustment; close application; restart; load saved data; compare values/history.	Product data and stock movement history remain available and consistent after restart.	UC01
TC-12	Input validation and typed error handling	Non-functional / Security	NFR02, SEC-03, VFR02	Application running; test invalid fields available.	Enter empty fields, invalid numeric values, negative quantities and invalid stock quantities; observe errors and recovery.	Invalid input is rejected before mutation; clear user-facing message is shown; application remains usable and expected errors do not crash the program.	UC02/UC07
TC-13	Authentication lockout and session gate	Security	FR01, SEC-01	Account exists; configured max attempts is 3 and lockout is 60 seconds.	Enter incorrect credentials repeatedly; attempt login while locked; after lockout, enter correct credentials; also attempt protected menu action without session.	Three consecutive failures trigger the configured lockout; protected commands require authentication; valid login after lockout succeeds and session is established.	UC01
TC-14	Ledger/file integrity on persistence failure	Security / Reliability	FR13, VFR07, NFR03, NFR04, SEC-04	Controlled test environment where a save/append failure can be simulated.	Trigger a file write/rename/ledger append failure during a stock adjustment; inspect product quantity, ledger and error.log.	Failed operation returns FILE_IO_ERROR; state is rolled back/restored as designed; no false successful ledger	UC07

						entry remains; error is logged.	
TC-15	Usability and normal-operation performance	Non-functional	NFR01, NFR07	Representative local dataset; normal supported environment.	Execute add, search, update, delete, quantity adjustment and retrieval operations repeatedly under normal usage; observe prompts/messages and responsiveness.	Menus/prompts/messages are clear; operations complete without unnecessary processing delays. No numeric performance threshold is imposed because the SRS defines none.	Multiple UCs
TC-16	Maintainability and module separation inspection	Non-functional / Inspection	NFR05	Source code and design baseline available.	Inspect layer boundaries, service/repository interfaces and dependency direction; verify business logic does not directly depend on file names/formats.	Presentation, services and repositories remain separated; module interfaces support independent testing; file I/O is concentrated in repositories/FileStore.	Architecture/design
TC-17	Portability and local-only operation	Non-functional	NFR06, NFR08	Compatible C++ environment on supported OS; no external DB/service available.	Build/run the application on the selected supported environment; disconnect/disable network access; execute core local operations.	Application builds/runs as a C++ local application using standard file I/O and core operations do not require network communication or an external database.	Architecture/deployment

16. Test Data and Boundary Conditions

Data Area	Representative Values / Conditions
Product ID	Valid unique ID; duplicate ID; empty ID; invalid/over-length ID.
Product name/category	Valid text; empty value; over-length value.
Unit price	0; positive value; negative value; non-numeric input.
Quantity	0; positive value; negative value; non-numeric input.
Stock-out	Quantity equal to available; quantity less than available; quantity greater than available; zero/negative quantity.
Low-stock threshold	Default threshold 10; product quantity exactly equal to threshold; quantity below threshold; quantity above threshold.
Authentication	Valid credentials; wrong password; unknown user; three consecutive failures; login during lockout.
Persistence	Fresh data directory; populated files; malformed line; write/rename failure.

